

## BALANCE–FIELD FRAMEWORK (BFG)

# The BFG Universal Reclosure Equation

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*A No-Retuning Unification from Level-0 Formation to Physical Matter,  
Abiogenesis, Organismic Integration, Conscious Subjectivity, and Qualitative  
Experience*

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### Scientific

### status

Independent theoretical whitepaper and falsifiable research program. The mathematical operator identities proved below are exact under their stated assumptions. The Universal Reclosure Equation is a new BFG unification construction built from the canonical closure machinery [1]. Stage identifications with actual physical, chemical, biological, neural, conscious, or phenomenal systems remain conditional or empirical until carrier-specific operationalizations pass prospective no-retuning tests. Synthetic simulations in this paper are executed demonstrations of the stated toy models; they are not empirical validation.

## Abstract

This whitepaper develops an independent BFG unification architecture in which the transition from Level-0 formation to physical closure, atomic and molecular organization, prebiotic chemistry, protocellular life, cellular and organismic integration, cognition, conscious subjectivity, phenomenal content, and selfhood is represented without introducing a new fundamental response law at each stage. The construction takes the canonical BFG positive master load  $Y$ , the exact neutral resolvent  $C_N(Y) = (I+Y)^{-1}$ , its complementary response  $B_N(Y) = Y(I+Y)^{-1}$ , the exact retained/complementary quadratic partition, recursive persistence, and the spectral formation gate as the mathematical spine [1,2]. The co-authored Level-0 foundation source line is retained for the pre-polar formation boundary [3]. The recursive conjugate-closure source line  $R_C = MRM^\dagger$ ,  $M = \sqrt{N}$  is preserved with explicit credit to Wendé and Leighton [4]. Existing BFG results provide fixed-point order, witness continuity, and conditional closure-to-next-trinity generation [5,6]. The present paper removes the remaining family of level-specific generativity maps by defining one Universal Reclosure Equation (URE),  $X_{n+1} = \mathcal{U}[X_n]$ , whose only admissible ingredients are upstream positive loads, the same neutral resolvent pair, a persistent-witness projector, the same positive closure geometry, and the spectral formation gate. Concrete stages are then defined as prospectively frozen invariant classes of iterates of  $\mathcal{U}$  rather than as new laws. This produces a single red thread: admissible difference  $\rightarrow$  neutral mediation  $\rightarrow$  exact retention/complement  $\rightarrow$  recursive persistence  $\rightarrow$  generative reclosure  $\rightarrow$  higher-order closure. The paper derives structural corollaries for physical geometry, nested matter, chemical and biological reclosure, organism formation, shared-neutral cognitive closure, and a non-circular closure-equivalence formulation of qualitative identity. Six deterministic numerical studies are executed: an exact two-eigenmode neutral partition surface, a spectral generativity normal form, an order-feedback stability surface, a shared-neutral integration surface, a preserved-differentiation surface, and a near-critical recursive trajectory. These calculations test internal mathematical consequences only. The strongest claim is therefore formal unification under no hidden retuning, not empirical proof that nature realizes every identified closure class.

**Keywords:** Balance–Field Framework; BFG; universal reclosure; Level-0; trinitarian formation; neutral resolvent; exact neutral elimination; recursive persistence; order-indexed mediation; no-retuning; abiogenesis; morphogenesis; organismic closure; consciousness; qualia; falsifiability.

## Contribution and source-line provenance

The present unification construction, the Universal Reclosure Equation, the no-hidden-retuning audit, the stage-predicate formalization, and the numerical synthesis are the responsibility of Marcel Theodor Wendé. Earlier BFG source lines are retained rather than absorbed into anonymous background.

- Marcel Theodor Wendé and Thomas W. Loker: co-authored Level-0 foundation source line used for pre-polar admissibility, determinate formation, and the Level-0 boundary [3].
- Marcel Theodor Wendé and Luke Kenneth Casson Leighton: recursive conjugate-closure source line, including the form  $R_C = MRM^\dagger$  with  $M = \sqrt{N}$  used in the persistence architecture [4].

## Reference and numbering convention

This whitepaper uses a single numerical reference sequence in order of first substantive appearance, with bracketed in-text citations and one numbered bibliography entry per source. This follows the professional sequential-numbering discipline used by major technical journals; the external bibliography was checked against publisher or authoritative indexing metadata where available on 7 September 2026. Equations, figures, and tables are each numbered sequentially throughout the manuscript.

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## 1. Purpose, scope, and the unification criterion

The central question of this paper is deliberately stronger than the question addressed by earlier cross-domain BFG formulations. It is not whether atoms, cells, organisms, and conscious systems can all be described with similar trinitarian language. It is whether those regimes can be placed on one calculational closure spine such that no stage acquires an independent fundamental nonlinear response law merely because the carrier changes. The canonical universal closure formulation already states this no-retuning requirement for physical sectors [1]. The present work extends the same discipline to the entire BFG closure hierarchy.

The unification criterion is therefore operational: after Level-0, every admitted stage must be obtained by reapplying the same master load construction, the same neutral resolvent family, the same persistence criterion, and the same spectral formation gate. Carrier-specific observables may change, and standard physics, chemistry, biology, and neuroscience remain necessary as empirical carrier descriptions. What is prohibited is a stage-specific BFG fundamental law inserted solely to make a desired higher stage appear.

$$\mathcal{T}_0 \rightarrow \Delta N_0 \rightarrow (P_1, N, P_2) \rightarrow K_0 \rightarrow \delta^* \rightarrow H\_C \rightarrow Y\_C \rightarrow \{C\_N, B\_N\} \rightarrow R\_C \rightarrow \mathcal{U} \rightarrow \mathcal{U}^2 \rightarrow \dots \quad (1)$$

Equation (1) is the red thread of the paper. The new symbol  $\mathcal{U}$  does not replace the canonical BFG closure machinery; it packages its repeated, no-retuning reapplication. The empirical question of whether real carriers occupy the resulting invariant classes is explicitly separated from the formal question of whether one mathematics is sufficient to define the hierarchy.

### Unification

### criterion

An alleged stage transition is not counted as unified if it requires a new sector-specific fundamental response function, a post-hoc role redefinition, a target-aware threshold, or a carrier map chosen only after the higher-level outcome is known.

## 2. Epistemic discipline and claim-state hierarchy

BFG already distinguishes definitions, axioms, theorems, conditional theorems, physical principles, model completions, spectral prescriptions, and empirical hypotheses [1,2]. This distinction is retained without compression. The new URE is a formal unification construction: it is a proposed rule for reapplying the canonical operator architecture, not an empirical law inferred from data.

**Table 1.** Claim-state discipline used throughout the paper.

| Class                           | Meaning                                                                                    |
|---------------------------------|--------------------------------------------------------------------------------------------|
| Definition                      | Introduces an object or equivalence class; true only by stipulation within the formalism.  |
| Axiom / inherited BFG structure | A structural assumption imported from the canonical corpus.                                |
| Theorem                         | Exact mathematical consequence under explicitly stated assumptions.                        |
| Conditional theorem             | Exact consequence after additional assumptions not yet derived upstream.                   |
| Physical principle              | A universal physical bridge constrained by no-retuning; LPCP is the canonical example [1]. |
| Unification construction        | New in this paper: a fixed operator recursion built only from inherited canonical objects. |
| Model completion                | A concrete representational or toy choice used to make a calculation executable.           |
| Spectral prescription           | A fixed readout from the master spectrum; numerical content must be computed.              |

|                      |                                                                                          |
|----------------------|------------------------------------------------------------------------------------------|
| Empirical hypothesis | Identification of a formal closure class with an observed physical or phenomenal regime. |
| Synthetic simulation | Numerically exact for the stated toy model; not empirical validation.                    |

The strongest permissible conclusion of the present paper is thus: one fixed BFG reclosure architecture can be written so that all later BFG stage labels are invariant predicates on its iterates. It is not permissible to convert that formal result into the statement that the universe has empirically been shown to follow the URE.

### 3. Canonical source architecture

#### 3.1 Level-0 trinitarian source

The canonical Level-0 formulation defines a finite-dimensional real Hilbert space  $V_0$  with three primitive admissibility roles: difference capacity  $\mathcal{D}_0$ , coherence capacity  $\mathfrak{c}_0$ , and neutral admissibility  $\mathfrak{N}_0$  [1,3].

$$\mathcal{T}_0 = (\mathcal{D}_0, \mathfrak{c}_0, \mathfrak{N}_0) \quad (2)$$

$$K_0 = \mathfrak{c}_0 + \mathfrak{N}_0 - \mathcal{D}_0 \quad (3)$$

$$\mathcal{A}_{\text{Adm}}(\delta) = \frac{1}{2}\langle \delta, K_0 \delta \rangle + \frac{1}{4}\|\delta\|^4, \quad \delta \in V_0 \quad (4)$$

A nontrivial isolated formation branch requires a simple negative minimum eigenvalue of  $K_0$  and a positive spectral gap.

$$\lambda_0 = \min \sigma(K_0) < 0, \quad \text{mult}(\lambda_0)=1 \quad (5)$$

$$\Delta_0 = \lambda_1 - \lambda_0 > 0 \quad (6)$$

$$\delta^{*\pm} = \pm\sqrt{-\lambda_0}e_0 \quad (7)$$

The quartic term makes the admissibility functional coercive; the Rayleigh–Ritz argument isolates the lowest eigenvector. The two minima are polarity-related. This is a mathematical formation result inside the stated real finite-dimensional model, not an empirical cosmological timestamp [1–3].

#### 3.2 Exact neutral elimination

Let  $A$  map a difference channel into a neutral mediation channel. The canonical closure formulation uses a quadratic neutral completion whose auxiliary neutral variable can be eliminated exactly [1].

$$\mathcal{A}_{\text{C}}(D, \nu) = \frac{1}{2}\|D - A^\dagger \nu\|^2 + \frac{1}{2}\|\nu\|^2 \quad (8)$$

$$Y = A^\dagger A \geq 0 \quad (9)$$

$$\nu^* = (I + AA^\dagger)^{-1}AD \quad (10)$$

$$C_{\text{N}}(Y) = (I + Y)^{-1} \quad (11)$$

$$B_{\text{N}}(Y) = Y(I + Y)^{-1} = I - C_{\text{N}}(Y) \quad (12)$$

$$C_{\text{N}}(Y) + B_{\text{N}}(Y) = I \quad (13)$$

$$0 < C_{\text{N}}(Y) \leq I, \quad 0 \leq B_{\text{N}}(Y) < I \quad (14)$$

$$\mathcal{A}_{\text{C}}^{\wedge \min}(D) = \frac{1}{2}\langle D, C_{\text{N}}(Y)D \rangle \quad (15)$$

$$\frac{1}{2}\|D\|^2 = \frac{1}{2}\langle D, C_{\text{N}}D \rangle + \frac{1}{2}\langle D, B_{\text{N}}D \rangle \quad (16)$$

Equations (9)–(16) are the central exact identities of the unification. They supply a bounded retained response, a bounded complementary response, and an exact quadratic partition. Standard positive-matrix analysis supports the spectral and perturbative facts used here [7,8].

### 3.3 Positive master closure geometry and spectral sectors

The canonical universal closure architecture lifts the local neutral pair into a positive master closure geometry [1].

$$B_C = D_{cov} c \quad (17)$$

$$H_C = B_C^\dagger W_N B_C \geq 0 \quad (18)$$

$$Y_C = L_C^\dagger H_C L_C \geq 0 \quad (19)$$

Sector identity is licensed only by upstream spectral gaps. If  $S_C$  is the relevant self-adjoint master sector operator and  $\Delta_a$  are disjoint spectral windows, then:

$$P_a = \mathbb{1}_{\{\Delta_a\}}(S_C), \quad P_a P_b = \delta_{ab} P_a, \quad \Sigma_a P_a = I \quad (20)$$

$$\beta_a = B_N^{\wedge\{1/2\}} P_a B_N^{\wedge\{1/2\}}, \quad \beta_a \geq 0, \quad \Sigma_a \beta_a = B_N \quad (21)$$

This positive partition is the exact no-double-counting backbone of later readouts. A claimed higher-stage channel that has no upstream gap or invariant distinction is not permitted to become a new fundamental BFG sector simply by naming it.

### 3.4 Recursive persistence and source-line continuity

The recursive conjugate-closure source line is preserved explicitly. Its conjugate form is used in the persistence architecture [4]. The canonical universal architecture uses a protected persistence direction plus a stable complement [1].

$$R_C = M U_\Omega C_E M^\dagger, \quad M = \sqrt{N_R} \quad (22)$$

$$R_C e_T = e_T \quad (23)$$

$$\|R_C|_{\{E_s\}}\|_L \leq r < 1 \quad (24)$$

$$P_T = \text{RieszProj}_{\{\lambda=1\}}(Dc_{\text{phys}}) \quad (25)$$

For  $s = ae_T + s_s$ , the stable component contracts at least geometrically, so the full trajectory remains power bounded. This motivates the carrier-neutral witness notion used below. The mathematical distinction between eigenvalue information and robust subspace information follows standard perturbation theory [7,8].

## 4. Order-indexed neutral mediation

A prior BFG formulation separates the inherited source ratio  $\Phi_{\text{src},n}$  from the realized local order state  $\varphi_n$ , the neutral relation, the operator realization  $\hat{N}_n[\varphi_n]$ , and an observable scalar response  $\nu_n$  [5]. This separation is retained because it prevents a single symbol from carrying incompatible roles.

$$\Phi_{\text{src},n} = (E_{\text{exp},n} I_n) / (E_{\text{bind},n} I_{\{0,n\}}) \quad (26)$$

$$\varphi_n \nu_n(\varphi_n) = \Phi_{\text{src},n} \quad (27)$$

$$\varphi_n = F_n(\varphi_n) := \Phi_{\text{src},n} / \nu_n(\varphi_n) \quad (28)$$

If  $F_n$  maps a closed complete positive domain into itself and has Lipschitz constant  $L_n < 1$ , the Banach contraction principle gives a unique local fixed point and convergent Picard iteration [5,9]. Cross-level coupling with induced Lipschitz gain  $\kappa$  remains jointly contractive under the sufficient condition:

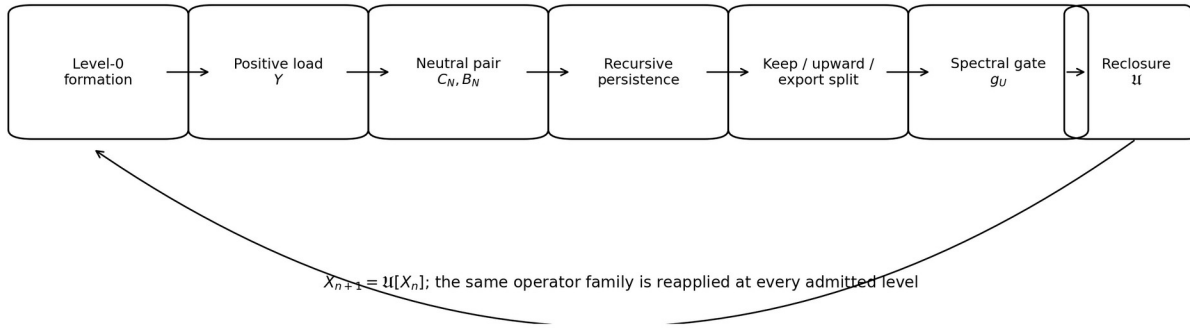
$$L + \kappa < 1, \quad L = \max_n L_n \quad (29)$$

The order-indexed extension also distinguishes structural level  $n$  from physical/developmental time  $t$ . This distinction is essential: the formal hierarchy is an order of closure precedence, not a claim that all physical systems literally pass through the same temporal sequence.

## 5. The Universal Reclosure Equation

### 5.1 Motivation

Prior BFG hierarchy formulations allowed level-specific transport and generativity maps  $G_n$  or  $G_n^V$  [5]. Those maps are appropriate for empirical operationalization, but they also create a formal place where hidden retuning can enter. The present paper therefore makes a stronger move: the generation map itself is no longer a free family. It is replaced by one universal operator recursion assembled solely from the canonical positive load, the exact neutral pair, the persistent-witness projector, and the spectral formation gate.



**Figure 1.** Universal reclosure architecture. The only iterated engine is the same sequence of formation, positive load, exact neutral resolution, persistence, retained/complement decomposition, spectral gating, and reclosure. The recurrence arrow denotes reapplication of the same operator family, not a new law at the next level.

### 5.2 Persistent witness projector

For any admitted recursive closure, define the power-bounded witness subspace as the set of directions that remain bounded under repeated recursive transport. In finite-dimensional models its robust spectral realization is represented by a Riesz projector around the protected persistence spectrum.

$$\mathcal{W}_n = \{z: \sup_{k \geq 0} \|R_{\{C,n\}}^k z\| < \infty\} \quad (30)$$

$$P_{\{\mathcal{W}_n\}} = \text{RieszProj}_{\{\Sigma_p\}}(R_{\{C,n\}}) \quad (31)$$

The exact choice of contour  $\Sigma_p$  is fixed prospectively around the protected spectral block. A post-hoc contour that is moved after observing a desired higher-stage outcome would violate no-retuning.

### 5.3 Canonical keep, upward, and export decomposition

The URE does not introduce an independent biological or cognitive information-gating operator. Instead it derives three operational roles from the same neutral pair and the same witness projector.

$$D_n^{\{\text{keep}\}} = P_{\{\mathcal{W}_n\}} C_N(Y_n) D_n \quad (32)$$

$$D_n^{\{\uparrow\}} = P_{\{\mathcal{W}_n\}} B_N(Y_n) D_n \quad (33)$$

$$D_n^{\{\text{out}\}} = (I - P_{\{\mathcal{W}_n\}}) B_N(Y_n) D_n \quad (34)$$

The three terms have different roles.  $D_n^{\text{keep}}$  is retained inside the current persistent closure;  $D_n^{\uparrow}$  is complementary structure that is nevertheless carried by a persistent witness and is therefore available to participate in a higher relation;  $D_n^{\text{out}}$  is complementary structure that does not belong to the declared persistent witness block and is exported from the higher-generation packet. These are operator-defined roles, not new substances.

#### 5.4 Next-level polarity packet and master load

The next-level candidate source is the direct-sum polarity packet formed from retained and persistent-complementary components. Direct sum is used here as a carrier-neutral bookkeeping operation; it does not assert that physical carriers literally double their microscopic dimension.

$$\forall D_{n+1} = D_n^{\text{keep}} \oplus D_n^{\uparrow} \quad (35)$$

$$Y_{n+1} = L_C \dagger H_C[\forall D_{n+1}] L_C \quad (36)$$

The functional form of  $H_C$  and  $L_C$  is not replaced by a level-specific Hessian. The same positive master closure construction is reevaluated on the candidate packet. This is the central no-retuning move.

#### 5.5 Universal spectral generativity gate

The canonical recursive spectral gate admits generation only if the next-level formation operator has a negative minimum eigenvalue [1]. The URE adopts this gate globally.

$$g_U(n) = \mathbb{1}_{\{\lambda_{\min}(K_{n+1}) < 0\}} \quad (37)$$

$$N_* = \min\{n: \lambda_{\min}(K_{n+1}) \geq 0\} \quad (38)$$

The gate is therefore not 'life-specific', 'atom-specific', or 'consciousness-specific'. It is the same spectral formation criterion at every admitted reclosure. A carrier may stop generating higher closure while continuing to persist and export; terminality is not annihilation.

#### 5.6 Universal Reclosure Equation

With the preceding objects fixed, the candidate higher-level retained state is obtained by applying the same neutral resolvent to the candidate polarity packet when and only when the common spectral gate passes.

$$X_{n+1} = \mathcal{U}[X_n] := g_U(n) C_N(Y_{n+1}) [D_n^{\text{keep}} \oplus D_n^{\uparrow}] \quad (39)$$

$$X_n = \mathcal{U}^n[X_0] \quad \text{for every admitted level before } N_* \quad (40)$$

$$g_U(n)=0 \Rightarrow \text{no higher BFG closure is admitted at } n+1 \quad (41)$$

**Status** of **the** **URE**  
Equations (36)–(38) are the new unification construction of this paper. They are built only from canonical BFG objects but the assertion that this reapplication is the correct universal cross-carrier generative rule is a theorem candidate / unification postulate, not an established fact of nature.

#### 5.7 No-Hidden-Retuning theorem candidate

The new paper can now state the strong universality condition precisely. Let  $\mathcal{P}_n$  be any stage-specific readout. A stage is formally unified only if  $\mathcal{P}_n$  acts on an iterate of  $\mathcal{U}$  without modifying the generating operator family.

$$\mathcal{P}_n: X_n \mapsto \text{observable stage readout}, \quad X_n = \mathcal{U}^n[X_0] \quad (42)$$

$$\mathcal{U}_{\text{atom}} = \mathcal{U}_{\text{bio}} = \mathcal{U}_{\text{cog}} = \mathcal{U}_{\text{phen}} = \mathcal{U} \quad (43)$$

Conditional theorem candidate. If (i) the Level-0 formation branch exists; (ii) every admitted iterate preserves positive master geometry; (iii) the persistent witness projector is fixed by upstream spectral structure; (iv) the same  $K$ ,  $H_C$ ,  $L_C$ ,  $C_N$ ,  $B_N$ , and persistence construction are used at every iterate; and (v) stage labels are fixed predicates on the

iterates rather than generators of the iterates, then no downstream stage requires an independent fundamental BFG response law. The proof is by construction: all dynamical generation is carried by the single map  $\mathcal{U}$ ; stage-specific objects appear only after the iterate exists.

## 6. What it means for a concrete stage to 'follow'

A conceptual difficulty must be removed at this point. A universal operator can generate a sequence of mathematical states, but the words atom, cell, organism, or conscious subject are empirical carrier labels. This paper therefore defines each concrete stage through a prospectively frozen invariant predicate  $S_j(X)$ . The operator generates  $X$ ; the predicate classifies  $X$ . The predicate does not feed back into  $\mathcal{U}$ .

$$\text{Stage}_j(X_n) = 1 \Leftrightarrow S_j(X_n) \text{ satisfies its frozen invariant conditions} \quad (44)$$

$$\text{Stage}_j \text{ is a readout predicate, not a term in } \mathcal{U} \quad (45)$$

This distinction allows all stages to follow from one generating equation without pretending that a name is a derivation. A real hydrogen atom is identified with the atomic closure class only if measured quantum/electromagnetic carrier data satisfy the frozen BFG atomic predicate. Likewise, a real conscious state is identified with the conscious closure class only if the shared-neutral, differentiation, self-indexing, and bounded-recursion conditions survive empirical tests [10].

## 7. Physical closure from the same architecture

### 7.1 Local Physical Closure Principle

The Local Physical Closure Principle (LPCP) states that once an admissible closure enters a smooth, reversible, locally geometric regime, local propagation is the continuum limit of covariantly transported neighboring closure differences, using the same positive neutral stiffness that controls admissibility [1]. LPCP is a physical principle, not a Level-0 theorem.

$$\Delta_{\{\varepsilon, \nu\}} q(x) = U_C(x, x + \varepsilon \nu) q(x + \varepsilon \nu) - q(x) \quad (46)$$

$$\Delta_{\{\varepsilon, \nu\}} q / \varepsilon = \nabla_\nu^\wedge C q + O(\varepsilon) \quad (47)$$

$$Z_C = H_C^{\{1/2\}} C_N(Y_C[q_*]) H_C^{\{1/2\}} > 0 \quad (48)$$

$$S_{LPC}^{\{(2)\}}[q] = \mathcal{K}_0 \int d^4x \sqrt{-g} [-\frac{1}{2} g^{\{\mu\nu\}} \langle \nabla_\mu^\wedge C q, Z_C \nabla_\nu^\wedge C q \rangle - V_C^{\{(2)\}}(q)] \quad (49)$$

$$V_C(q) = \frac{1}{2} \langle D_C(q), C_N(Y_C(q)) D_C(q) \rangle \quad (50)$$

The local response Hessian follows from derivatives of the same resolvent, rather than from an independent sector response.

$$\partial_A C = -C Y_{,A} C \quad (51)$$

$$\partial_A \partial_B C = C Y_{,A} C Y_{,B} C + C Y_{,B} C Y_{,A} C - C Y_{,AB} C \quad (52)$$

In the minimal single-metric, two-derivative branch, the principal symbol of every nondegenerate fundamental massless block is proportional to one metric null form. Under additional standard assumptions, the geometric gauge block is related to Fierz–Pauli massless spin-2 and the Einstein infrared branch [11–15]. These are conditional physical consequences, not additional BFG fundamental laws.

$$\sigma_2(\Gamma_a)(x, k) = Z_a g^{\{\mu\nu\}} k_\mu k_\nu P_a \quad (53)$$

$$g^{\{\mu\nu\}} k_\mu k_\nu = 0 \quad (54)$$

## 7.2 Physical-stage predicate

A URE iterate is classified as entering the local physical-closure branch only when it supports a smooth local carrier, nondegenerate soldering, positive closure stiffness, reversible local transport, and a stable characteristic structure. These requirements are readouts of the iterate plus LPCP. They do not modify  $\mathcal{U}$ .

### Physical-stage

status

**The existence of an admissible local geometric branch is conditional. The identification of the resulting metric and spectrum with observed spacetime and gravity remains empirical. The standard field-theory literature constrains the carrier interpretation [11–15]; it does not independently derive BFG.**

## 8. Matter: nested nuclear, atomic, and molecular closure

The atomic stage is constrained as a nested neutral-mediated closure while its stability grammar is explicitly not treated as a replacement for quantum mechanics, QED, nuclear physics, or chemistry [16]. In the present unification, atomic language becomes a stage predicate on  $\mathcal{U}$  rather than a separate source of fundamental mathematics.

### 8.1 Atomic predicate

A physical URE iterate qualifies as an atomic closure candidate when: (i) it contains an internally bound, spectrally separated persistent block; (ii) distinguishable charge-related carrier roles remain noncollapsed; (iii) the retained block is stable under recursive transport; and (iv) nonintegrable excitation can leave through the universal complementary/export channel. Standard quantum operators determine whether a real carrier actually realizes these conditions; the BFG response family is not changed.

$$\text{Atomic}(X_n)=1 \Rightarrow \text{gap\_spec}>0, \sup_k \|R_C^k X_n\|<\infty, D_n^{\text{out}} \text{ permitted} \quad (55)$$

This is intentionally a structural implication rather than a derivation of the hydrogen spectrum from a scalar toy load. The no-retuning claim is that the BFG closure law is unchanged; standard physics supplies the carrier Hamiltonian and measured spectral data.

### 8.2 Molecular and chemical reclosure

Once lower closures are stable, more than one may enter the retained/complement polarity packet of a higher iterate. A molecular candidate is therefore a nonfactorizable higher closure of previously stable atomic candidates. The carrier change is constitutive: atoms remain identifiable subclosures while a new relational spectrum appears.

$$C_{\text{mol}} \neq C_{\text{atom},1} \oplus C_{\text{atom},2} \quad \text{when the higher retained response is nonfactorizable} \quad (56)$$

The distinction between constitutive nesting and temporal generation is important. The hierarchy expresses closure precedence. It does not imply that every molecule must literally replay Level-0 or atomic formation as a temporal developmental sequence.

## 9. From chemistry to abiogenesis

The molecular-to-biological transition is treated as a falsifiable generativity problem rather than an adjacent pair of labels [5]. External origin-of-life research shows that several carrier-level ingredients are experimentally tractable: activated pyrimidine ribonucleotides can arise under prebiotically plausible chemistry [17], common precursor chemistry can feed RNA, amino-acid, and lipid building blocks [18], template-directed polymer synthesis can occur in model protocells [19,20], protocell populations can exhibit competition [21], and recent autocatalytic systems can couple reaction dynamics to compartment growth and reproduction-like behavior [22]. These results constrain carrier plausibility; none validates BFG.

## 9.1 Prebiotic closure as URE readout

Under the URE, reaction-network organization, compartment persistence, throughput, renewal, heredity, and population persistence are not additional fundamental terms. They are observable signatures of a particular higher-order closure class. The generative law remains Equation (36).

$$C_{\text{prebiotic}} = \text{Stage}_{\text{prebiotic}}(\mathcal{U}^n[X_0]) \quad (57)$$

$$C_{\text{proto}} = \text{Stage}_{\text{proto}}(\mathcal{U}^{\{n+1\}}[X_0]) \quad \text{only if } g_U(n)=1 \quad (58)$$

For empirical testing, an existing multicomponent abiogenesis observation vector remains useful: boundary persistence  $q_B$ , flux/export  $q_F$ , renewal  $q_R$ , heredity  $q_H$ , cross-sector coupling  $q_C$ , and differential persistence/selection  $q_S$  [5]. In the unification construction these quantities diagnose whether an iterate belongs to the protocellular class; they no longer constitute a separate fundamental gate.

$$\vec{q}_{\text{abi}} = (q_B, q_F, q_R, q_H, q_C, q_S) \quad (59)$$

A strong empirical test compares the full frozen BFG readout against capacity-matched hierarchy-only, compartment-only, heredity-only, and shuffled-coupling nulls. The repeated-cycle requirement remains: a one-off vesicle or isolated copying event is not enough to identify biological closure [5,23].

## 10. Cellular, multicellular, and organismic reclosure

### 10.1 Cellular closure

The established BFG stage ordering places molecular, sequence, informational, replicative, error-correction, metabolic, boundary, protocellular, projection, readout, feedback, network, field, attractor, differentiation, tissue, organ, organ-system, and organismic regimes in precedence order [24]. In the present paper these are not separate laws. They are successive invariant signatures that may appear along the same URE orbit if the common spectral gate continues to pass.

A cell is identified when a protocellular closure becomes recursively self-maintaining across material turnover while retaining a boundary witness, throughput/export, informational renewal, and a return structure that preserves closure-relevant invariants. The URE itself contains no gene-specific or membrane-specific response function; those are carrier mechanisms to be tested against the frozen structural predicate.

### 10.2 Multicellular projection and morphogenesis

A prior organismic formulation treats local cellular closures as projecting into a form/field domain, being read out, differentiated, and reintegrated into a global closure [25]. Classical developmental biology provides independent mechanisms—reaction–diffusion, positional information, morphogen interpretation, gene-regulatory organization, mechanics, and bioelectric pattern control [26–30]. BFG does not replace these mechanisms. The unification claim is that they can be evaluated as carrier realizations of a common closure transition.

$$C_{\text{global}} = \text{Stage}_{\text{organism}}(\mathcal{U}^m[\{C_i\}]) \quad (60)$$

A multicellular mass does not qualify merely because cell number increases. The organismic predicate requires persistent global integration, preserved local differentiation, admissible projection/readout, regulated export, and failure under the appropriate ablations. The distinction cell growth  $\neq$  organism formation is retained [25].

### 10.3 Organismic closure

The organismic regime is the first biological stage in which many lower closures must remain locally viable while being globally constrained by a higher closure. The URE therefore interprets organism formation as many-to-one reclosure under one fixed generator. This is compatible with systems-biology and autopoietic descriptions of organized living systems [31,32], but the BFG-specific neutral architecture must add prospective explanatory or predictive value to survive null-model comparison.

## 11. Sensorimotor, cognitive, and memory closure

After organismic closure, the BFG biological ordering introduces sensory, integration, valence, action, learning, memory, self-model, perspective, and identity regimes [24]. The unification paper treats each as a structural refinement of the same recursive closure rather than as a new fundamental domain.

**Table 2.** Cognitive-stage predicates as readouts of one URE orbit.

| Stage               | Frozen structural signature                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------|
| Sensory closure     | Externally induced differences enter a retained internal channel while remaining discriminable.            |
| Integration closure | Multiple channels become nonfactorizably coupled through one shared neutral load.                          |
| Valence closure     | Alternative states acquire different closure-persistence/export gradients; empirical mapping remains open. |
| Action closure      | Export changes the external carrier and returns as a new admissible input.                                 |
| Learning closure    | Later reclosure changes prospectively measurable transition behavior after perturbation.                   |
| Memory closure      | A declared witness remains recoverable across temporally separated recursive cycles.                       |
| Self-model closure  | The closure contains a stable internal model/readout of its own persistent state.                          |
| Perspective closure | World-related and self-related sectors remain distinct but mutually mediated.                              |
| Identity closure    | A persistent invariant subspace survives while transverse degrees of freedom adapt.                        |

## 12. Thermodynamic maintenance as a readout, not a second consciousness law

A prior thermodynamic model introduced explicit entropy, information, and energy-balance equations [33]. In the present unification those equations are not used as a second fundamental definition of consciousness. Instead, thermodynamic quantities are read from a physical carrier that realizes the universal retained/complement partition.

For a normalized carrier state  $D$ , define retained and complementary quadratic fractions:

$$r_N(D) = \langle D, C_N(Y)D \rangle / \langle D, D \rangle \quad (61)$$

$$b_N(D) = \langle D, B_N(Y)D \rangle / \langle D, D \rangle = 1 - r_N(D) \quad (62)$$

A physical model may then map these fractions into bound energy, exported heat, chemical work, or information-processing costs. Landauer's result constrains the energetic cost of logically irreversible information operations [34], but it does not imply that consciousness is simply entropy reduction.

$$E_{\text{bind}} = \mathcal{R}_E^{\{\text{bind}\}}[r_N, \text{carrier}] \quad (63)$$

$$E_{\text{exp}} = \mathcal{R}_E^{\{\text{out}\}}[b_N, \text{carrier}] \quad (64)$$

The readout maps  $\mathcal{R}_E$  are empirical constitutive maps, not new fundamental BFG dynamics: they do not feed back into  $\mathcal{U}$  unless their measured carrier state changes  $Y$  in the already-declared upstream way. This preserves the distinction between a universal closure law and domain-specific thermodynamic units.

### 13. Conscious subjectivity as a shared-neutral URE class

The shared-neutral subjectivity formulation is especially compatible with the unification because its central mediation map already generates the same positive-load/resolvent family [10]. Contemporary consciousness science contains several non-equivalent theories and emphasizes the need for experimentally discriminable predictions [35–38]. The 2025 adversarial collaboration between GNWT and IIT provides a particularly relevant methodological benchmark for preregistered theory comparison [39].

$$Y_{\varphi} = M_{\varphi} \dagger M_{\varphi} \geq 0 \quad (65)$$

$$C_{\varphi} = (I + Y_{\varphi})^{-1} \quad (66)$$

$$Q_{\varphi} = Y_{\varphi} (I + Y_{\varphi})^{-1} = I - C_{\varphi} \quad (67)$$

Relative to a fixed channel decomposition with projectors  $P_i$ , exact channel factorization is equivalent for  $Y_{\varphi}$  and  $C_{\varphi}$ . Shared-neutral integration therefore requires nonzero off-diagonal response, while differentiation is protected by the smallest singular value.

$$Y_{\{ij\}}(\varphi) = P_i Y_{\varphi} P_j, \quad C_{\{ij\}}(\varphi) = P_i C_{\varphi} P_j \quad (68)$$

$$\Xi_{\varphi} = [\Sigma_{\{i \neq j\}} \|P_i C_{\varphi} P_j\|^2_{\text{HS}}] / \|C_{\varphi}\|^2_{\text{HS}} \quad (69)$$

$$\Delta_{\varphi} = \sigma_{\min}(C_{\varphi}|_{\{H_{\text{act}}\}}) = 1/[1 + \lambda_{\max}(Y_{\varphi}|_{\{H_{\text{act}}\}})] \quad (70)$$

A conscious-subjectivity candidate additionally requires world/self relation, self-indexing, and bounded recursion. These are not collapsed into one scalar. The empirical identification remains a hypothesis [10].

$$K_{\varphi} = R_{\varphi} C_{\varphi} \quad (71)$$

$$K_{\varphi} \dagger L K_{\varphi} \leq L \quad (72)$$

$$\sup_n \|K_{\varphi}^n x\|_L < \infty \quad (73)$$

Perturbational-complexity and near-criticality research provide external benchmarks for integration-plus-differentiation and bounded dynamical regimes [40,41]. BFG does not identify its variables directly with PCI or with a generic criticality index; such mappings must be preregistered.

### 14. Phenomenal content and qualitative identity

A prior qualitative-content formulation treats red as neither the wavelength, a neural label, nor a semantic token, but as the intrinsic invariant of a visually mediated, self-owned recursive closure [42]. It expresses this with an abstract invariant extractor  $\text{Inv}_G$  while explicitly acknowledging that a unique non-circular empirical estimator remains open. The unification paper removes  $\text{Inv}_G$  as a new free fundamental operator and replaces it with an equivalence class determined by invariants already available from the URE.

#### 14.1 BFG closure equivalence

Let  $\mathcal{G}_{\text{adm}}$  be the set of admissible transformations that preserve the declared witness spectrum, the neutral-response spectra, the local order state within tolerance, and the restricted recursive transport. Define:

$$X \sim_{\text{BFG}} X' \Leftrightarrow \exists G \in \mathcal{G}_{\text{adm}}: X' = GX \text{ and } \mathbb{I}_U(X') = \mathbb{I}_U(X) \quad (74)$$

$$\mathbb{I}_U(X) := \{\text{Spec } C_N(Y_X), \text{Spec } B_N(Y_X), \varphi_X, \mathcal{W}_X, R_C[\{\mathcal{W}_X\}]\} \quad (75)$$

$$Q[X] := [X]_{\sim_{\text{BFG}}} \quad (76)$$

The formal qualitative identity is therefore an equivalence class of the closure geometry, not an extra postulated substance. For two content closures:

$$Q_{\text{red}} \neq Q_{\text{blue}} \Leftrightarrow [C_{\text{red}}]_{\{\sim_{\text{BFG}}\}} \neq [C_{\text{blue}}]_{\{\sim_{\text{BFG}}\}} \quad (77)$$

This construction closes the formal operator gap only at the mathematical level. The empirical claim that a particular equivalence class is phenomenal redness still requires content-specific neural and behavioral tests. If closure invariants fail to distinguish red from blue after controlling local activation, or if the equivalence relation can only be defined using the target report itself, the BFG qualia claim is weakened or fails [42].

## 14.2 Ownership and selfhood

The same qualitative-content formulation distinguishes qualitative appearance from 'my appearance' [42]. In URE terms, ownership is a higher reclosure linking the content-equivalence class to the persistent identity witness. The content need not be fundamentally altered; what changes is its position in the higher closure.

$$\text{SelfOwned}(Q_i)=1 \Leftrightarrow Q_i \text{ is bidirectionally reachable from the persistent identity witness } \mathcal{W}_{\text{id}} \quad (78)$$

External work on bodily self-consciousness supports the broader empirical fact that ownership, self-location, perspective, and multisensory self-processing can dissociate [43]. It does not validate the BFG selfhood operator; it supplies independent phenomena against which the closure architecture can be tested.

## 15. The complete cosmic precedence map

The word cosmic in this paper denotes structural precedence, not a literal claim about a single historical timeline. Physical cosmology, chemical assembly, biological development, and cognitive development occur on different carrier timescales. The URE orders them by the first appearance of invariant closure capabilities. The map below is therefore a formal precedence ladder.

**Table 3.** Universal Reclosure stage map in structural precedence order.

| Index | Closure class                   | URE-derived structural predicate                                                                  |
|-------|---------------------------------|---------------------------------------------------------------------------------------------------|
| 0     | Level-0 admissibility           | Difference capacity, coherence capacity, neutral admissibility; isolated negative formation mode. |
| 1     | Primitive formed closure        | $\delta^*$ exists and the positive master load is finite.                                         |
| 2     | Neutral-resolved closure        | Exact $C\_N/B\_N$ partition is defined.                                                           |
| 3     | Persistent closure              | Protected witness block is power bounded.                                                         |
| 4     | Local physical closure          | Smooth reversible local carrier satisfies LPCP conditions.                                        |
| 5     | Bound physical subclosure       | Spectrally isolated retained block with admissible export.                                        |
| 6     | Nuclear/atomic class            | Nested physical bound closure with discrete persistent spectral structure.                        |
| 7     | Molecular class                 | Nonfactorizable higher closure of stable atomic subclosures.                                      |
| 8     | Macromolecular / sequence class | Persistent relational ordering becomes a witness across reclosure.                                |
| 9     | Informational class             | Alternative sequence states remain discriminable and transportable.                               |
| 10    | Replication class               | Closure-relevant witness is reproduced into an additional same-level closure.                     |
| 11    | Error-correction class          | Perturbed states return to the same closure-equivalence class.                                    |
| 12    | Metabolic throughput class      | Persistent retention coexists with regulated physical export/renewal.                             |

|    |                                              |                                                                                                            |
|----|----------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 13 | Boundary class                               | Stable inside/outside partition with selective exchange.                                                   |
| 14 | Protocellular class                          | Boundary, throughput, renewal, heredity, coupling, and repeated-cycle persistence jointly hold.            |
| 15 | Cellular class                               | Protocellular closure becomes recursively self-maintaining across turnover.                                |
| 16 | Projection class                             | Persistent complementary structure becomes available to other local closures.                              |
| 17 | Readout class                                | Projected structure enters another closure's neutral mediation.                                            |
| 18 | Feedback class                               | Projection and return form a bounded recursive loop.                                                       |
| 19 | Network / field class                        | Multiple same-level closures become nonfactorizably coupled.                                               |
| 20 | Attractor class                              | Perturbed states return to a stable closure basin.                                                         |
| 21 | Differentiation class                        | Distinct local attractors persist under one higher closure.                                                |
| 22 | Tissue class                                 | Many cell closures form a persistent spatial-functional higher closure.                                    |
| 23 | Organ class                                  | Specialized tissue closures reclose into a higher persistent function.                                     |
| 24 | Organ-system class                           | Multiple organs form an integrated higher closure.                                                         |
| 25 | Organism class                               | Global closure persists despite internal turnover and differentiation.                                     |
| 26 | Sensory class                                | External differences are retained as discriminable internal states.                                        |
| 27 | Integration class                            | Multiple channels share a nonfactorizable neutral load.                                                    |
| 28 | Valence class                                | Alternatives acquire distinct persistence/export gradients.                                                |
| 29 | Action class                                 | Export alters the environment and returns as structured input.                                             |
| 30 | Learning class                               | Future reclosure changes after prior perturbation/feedback.                                                |
| 31 | Memory class                                 | Witnesses persist across separated recursive cycles.                                                       |
| 32 | Self-model class                             | Closure contains a persistent internal model of its own state.                                             |
| 33 | Perspective class                            | World/self poles are distinct but jointly mediated.                                                        |
| 34 | Identity class                               | A persistent self-index survives adaptive transverse change.                                               |
| 35 | Conscious-subjectivity candidate             | Connected shared-neutral integration + differentiation + world/self + identity + bounded recursion.        |
| 36 | Phenomenal-unity candidate                   | Multiple conscious content closures occupy one globally connected differentiated field.                    |
| 37 | Qualitative-content class                    | Content is identified by a non-circular URE closure-equivalence class.                                     |
| 38 | Self-owned experience                        | Content is bidirectionally linked to the persistent identity witness.                                      |
| 39 | Narrative / intersubjective extension        | Higher closures incorporate temporally extended self-models and other-subject relations.                   |
| 40 | Knowledge / science / civilization extension | Collective closures preserve and transform shared witnesses; strongly empirical/speculative BFG extension. |

## 16. Executed numerical demonstrations

All simulations in this section were executed for the present whitepaper in Python 3.13.5 using NumPy 2.3.5 and Matplotlib 3.10.8. They are deterministic and dimensionless. No empirical dataset was fitted. The simulations are designed to test algebraic or stability consequences of the declared equations and to expose failure surfaces; they do not demonstrate that any physical, biological, or conscious system realizes BFG.

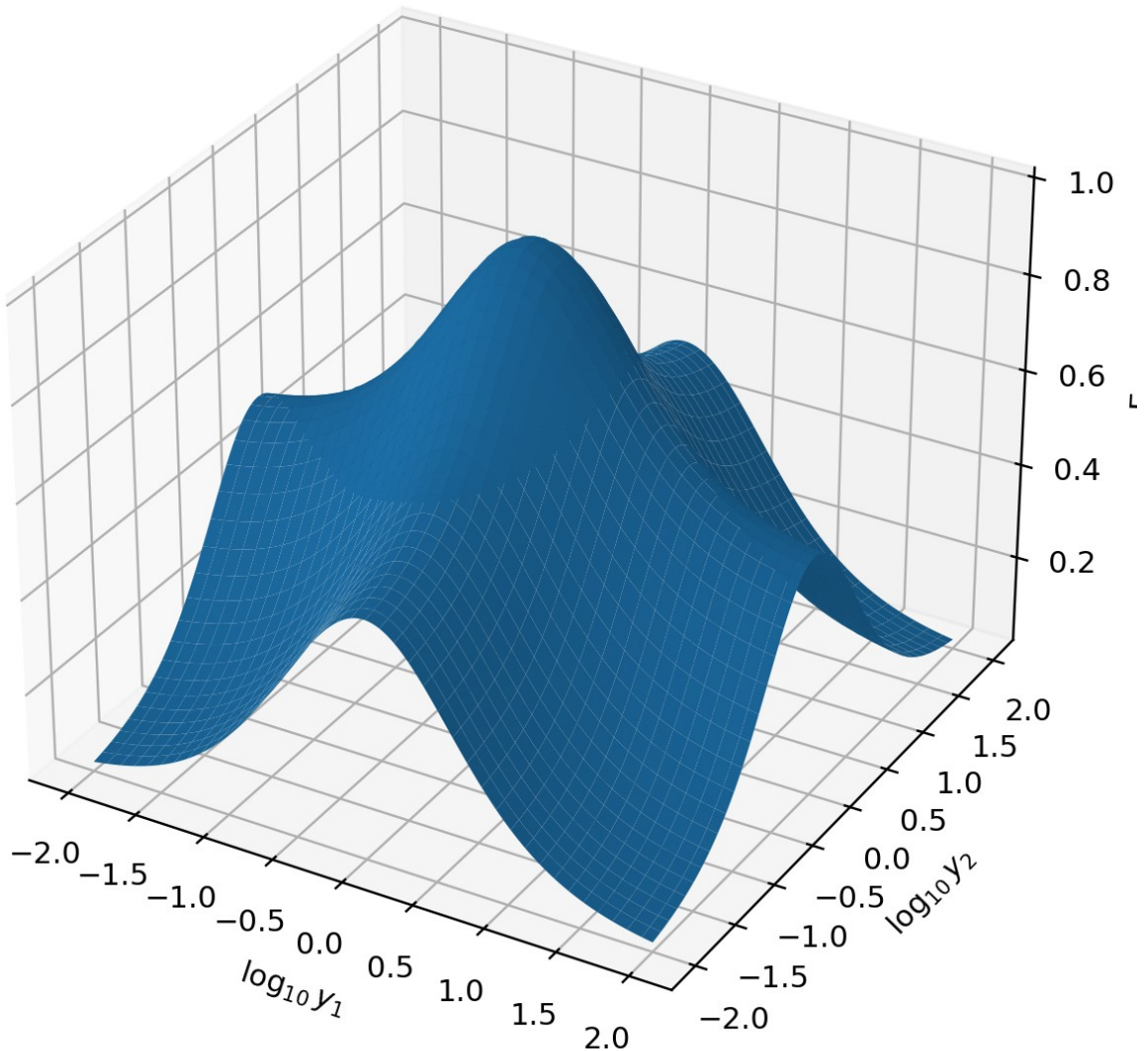
### 16.1 Exact neutral-partition surface

For two independent eigenloads  $y_1, y_2 \geq 0$ , define  $c_i = (1 + y_i)^{-1}$  and  $b_i = y_i / (1 + y_i)$ . The exact partition  $c_i + b_i = 1$  holds independently on each eigenmode. A symmetric diagnostic of simultaneous retained/complement availability is:

$$\Gamma_{rc}(y_1, y_2) = \frac{1}{2} [4c_1b_1 + 4c_2b_2], \quad 0 \leq \Gamma_{rc} \leq 1 \quad (79)$$

The numerical sweep used  $y_1, y_2 \in [10^{-2}, 10^2]$ . The maximum grid value was 0.999790 at  $y_1 \approx 0.9715$ ,  $y_2 \approx 0.9715$ ; the maximum absolute partition error was  $2.220e-16$ , consistent with floating-point roundoff. The analytic maximum  $\Gamma_{rc} = 1$  occurs at  $y_1 = y_2 = 1$ , where retained and complementary fractions are balanced mode by mode.

### Exact Neutral Partition: Mean Eigenmode Reclosure Availability



**Figure 2.** Executed 3D neutral-partition simulation. The surface is derived solely from the exact BFG resolvent pair. The ridge collapses to the analytic balanced maximum at  $y_1=y_2=1$ ; no sector-specific response function is used.

## 16.2 Spectral generativity normal form

The spectral gate depends only on the sign of the next-level minimum formation eigenvalue. To visualize this sign change without assigning a physical sector, a two-mode symmetric normal form was used:

$$K_{nf}(\delta, g) = [[1+\delta, -g], [-g, 1-\delta]] \quad (80)$$

$$\lambda_{\min}(K_{nf}) = 1 - \sqrt{(\delta^2 + g^2)} \quad (81)$$

$$g_U = 1 \Leftrightarrow \delta^2 + g^2 > 1 \quad (82)$$

A dense sweep over  $\delta \in [-1.6, 1.6]$  and  $g \in [0, 1.6]$  reproduced the analytic circular sign boundary exactly to grid resolution. 69.5% of the displayed rectangular parameter window lies in the negative-eigenvalue region; this percentage has no physical meaning and depends only on the chosen plotting window. The scientific content is the visible gate surface  $\lambda_{\min}=0$ .

### Spectral Generativity Gate: Two-Mode Normal Form

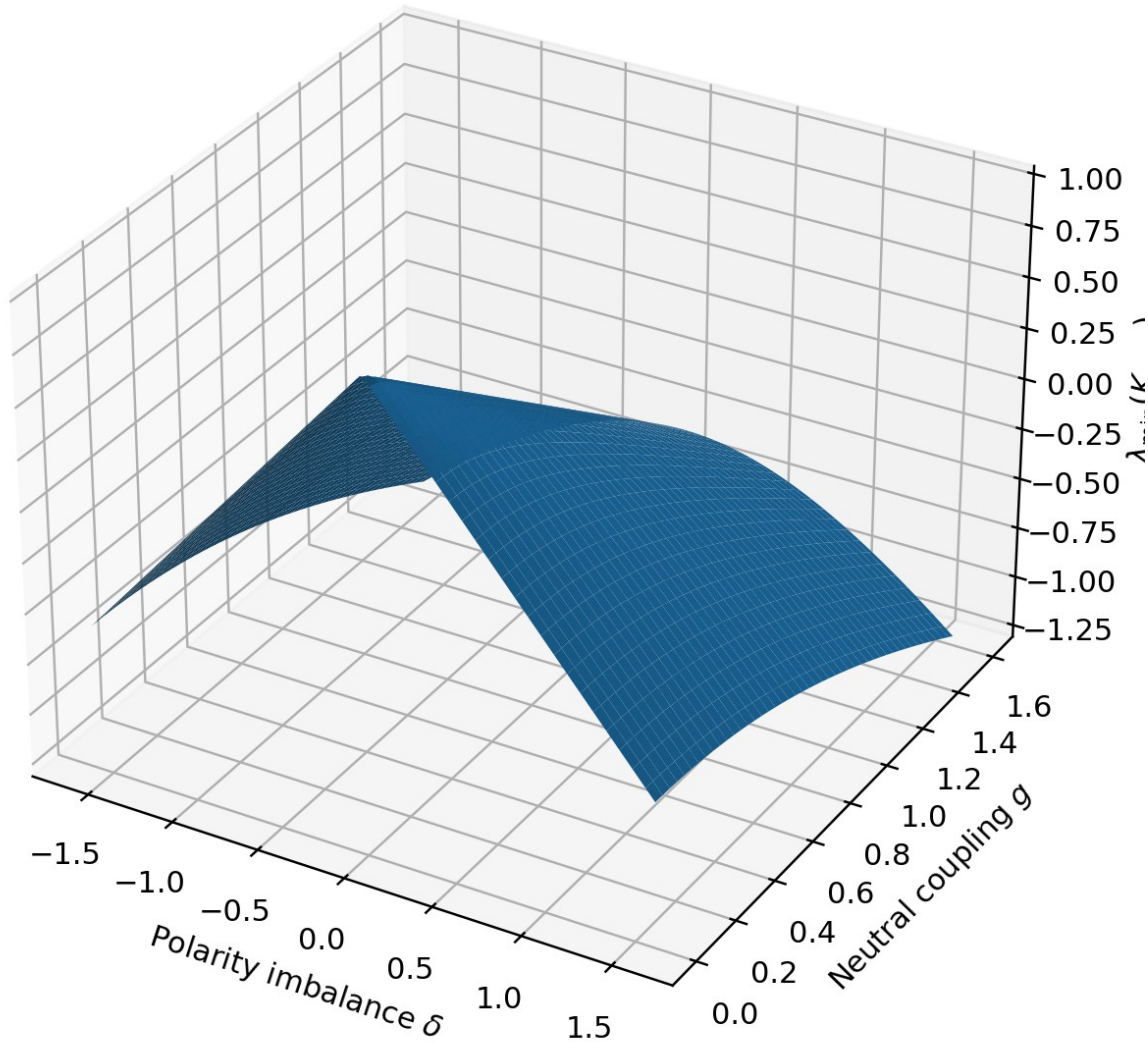


Figure 3. Executed 3D spectral-gate normal form. The surface illustrates the canonical rule  $g_U=1$  when the candidate next-level formation operator acquires a negative minimum eigenvalue. The matrix is a pedagogical normal form, not a carrier law.

### 16.3 Order-feedback stability corridor

The declared order-indexed toy Jacobian uses  $v(\varphi)=\exp[-4(\varphi-1)^2]$ ,  $\alpha=0.35$ ,  $\beta=0.25$ ,  $d=0.45$ , and  $c=0.08$  [5]. The present work re-executed the full  $(\varphi, B)$  spectral-radius sweep.

$$v(\varphi)=\exp[-4(\varphi-1)^2] \quad (83)$$

$$J(\varphi, B) = [[B + 0.35(1-v), 0.08v], [0.08v, 0.45 + 0.25(1-v)]] \quad (84)$$

At  $B=0.82$  the executed values are  $\rho(J) = 0.8366$  at  $\varphi=1.0$  and  $0.9897$  at  $\varphi=1.4$ . At  $B=0.90$  and  $\varphi=1.4$ ,  $\rho(J) = 1.0690$ , crossing the linear stability boundary. Across the displayed  $\varphi \in [0.45, 1.55]$ ,  $B \in [0.55, 0.98]$  grid, 78.9% of sampled points had  $\rho < 1$ . That fraction is a plotting statistic, not a universal prediction.

### Order-Indexed Stability Corridor

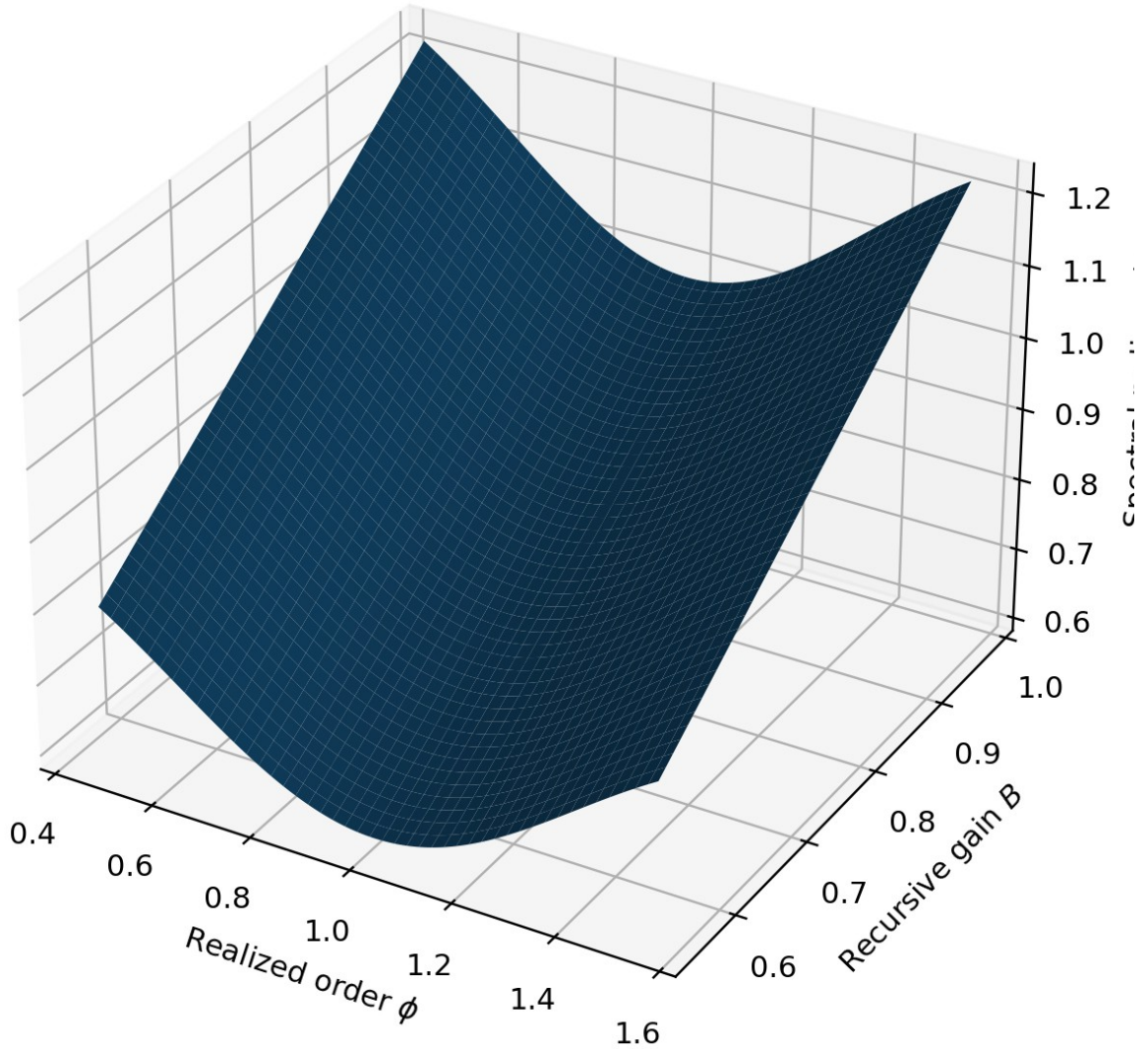


Figure 4. Executed 3D spectral-radius surface for the order-indexed interface toy model. The surface reproduces the finite joint stability corridor associated with the cited parameterization [5]. Linear spectral stability is necessary only for this fixed linear toy; nonnormal and time-varying systems require stronger power-bounded or Lyapunov analysis.

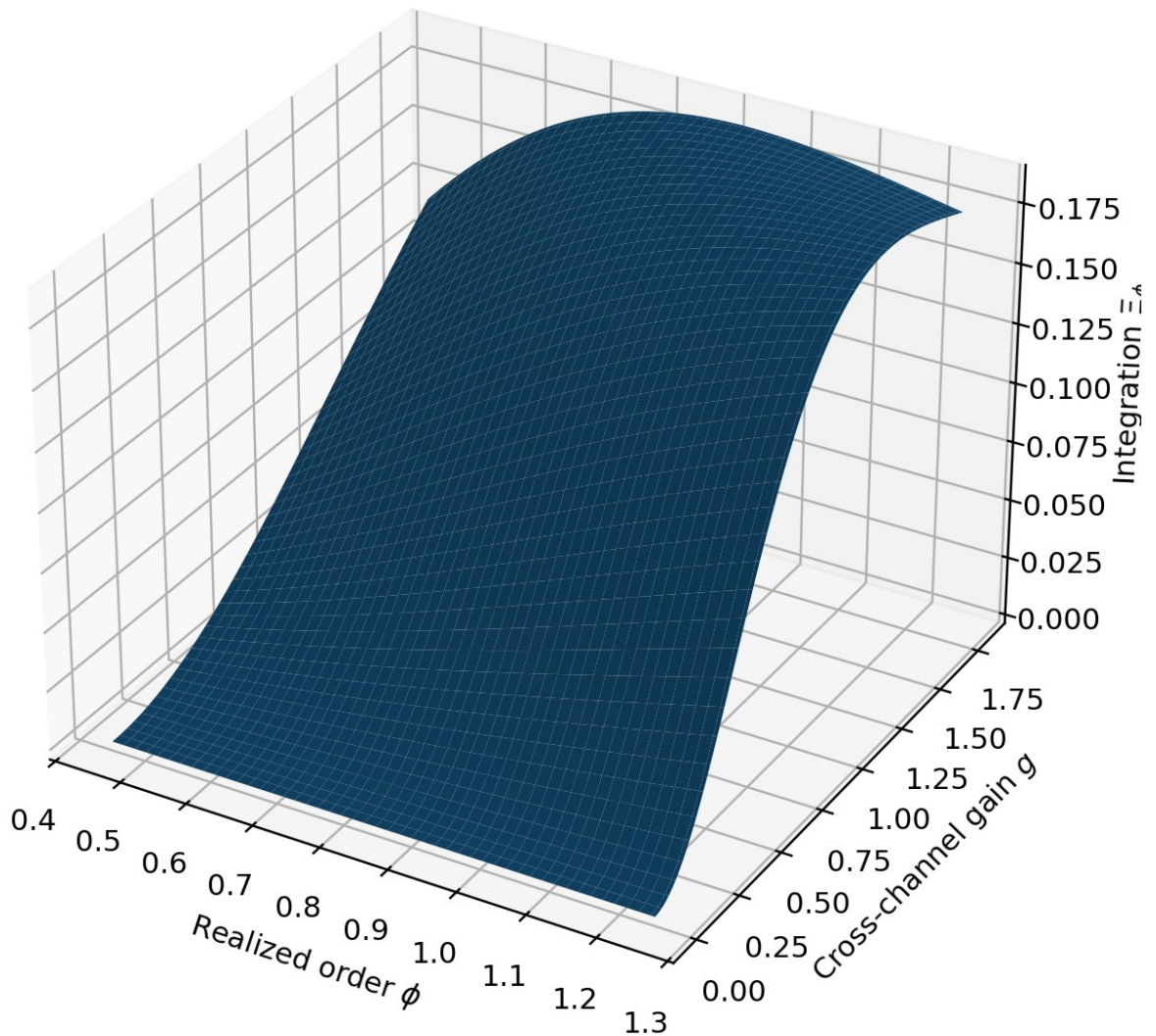
## 16.4 Shared-neutral cognitive integration

The five-channel cognitive simulation was re-executed from the previously defined matrices [10]. The mediation map uses  $D=\text{diag}(0.55,0.65,0.75,0.85,0.95)$ , a symmetric five-node ring with three weak chords normalized to spectral norm one, and  $M_\varphi=D+g\varphi W$ . With  $\Phi_{\text{src}}=1$  and  $v(\varphi)=1+0.45\varphi$ , the positive order fixed point is:

$$\varphi^*=[-1+\sqrt{(1+4a\Phi_{\text{src}})}]/(2a) \quad (85)$$

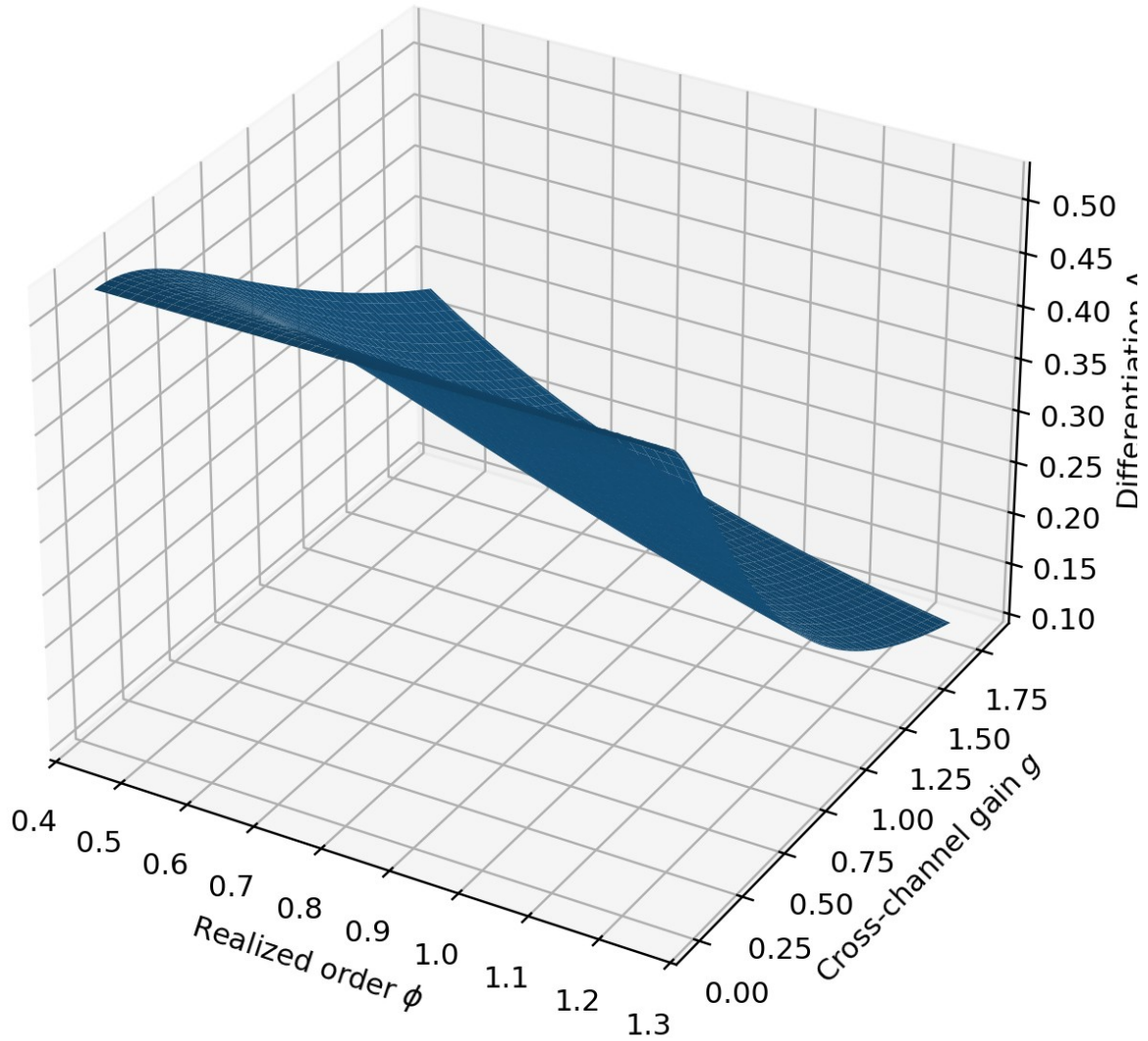
The executed value is  $\varphi^*=0.7481$ . At  $g=0,0.4,0.8,1.2,1.6$  the integration/differentiation pairs are respectively 0.0: (0.0000, 0.5256); 0.4: (0.0261, 0.4482); 0.8: (0.0819, 0.3442); 1.2: (0.1350, 0.2642); 1.6: (0.1697, 0.2061). These reproduce the previously defined trade-off [10]: coupling raises nonfactorizable integration while lowering the guaranteed differentiation margin.

### Shared-Neutral Integration Surface



**Figure 5.** Executed 3D shared-neutral integration surface  $\Xi_\varphi$ . Integration vanishes in the  $g=0$  ablation and grows as the common mediation map develops off-diagonal channel coupling. The plot is an internal mathematical demonstrator, not a consciousness measurement.

## Preserved-Differentiation Surface



**Figure 6.** Executed 3D differentiation surface  $\Delta_\phi = \sigma_{\min}(C_\phi)$  over the same parameter plane. Increasing shared load can erode the minimum distinguishability margin; this is why BFG does not identify maximal coupling with consciousness.

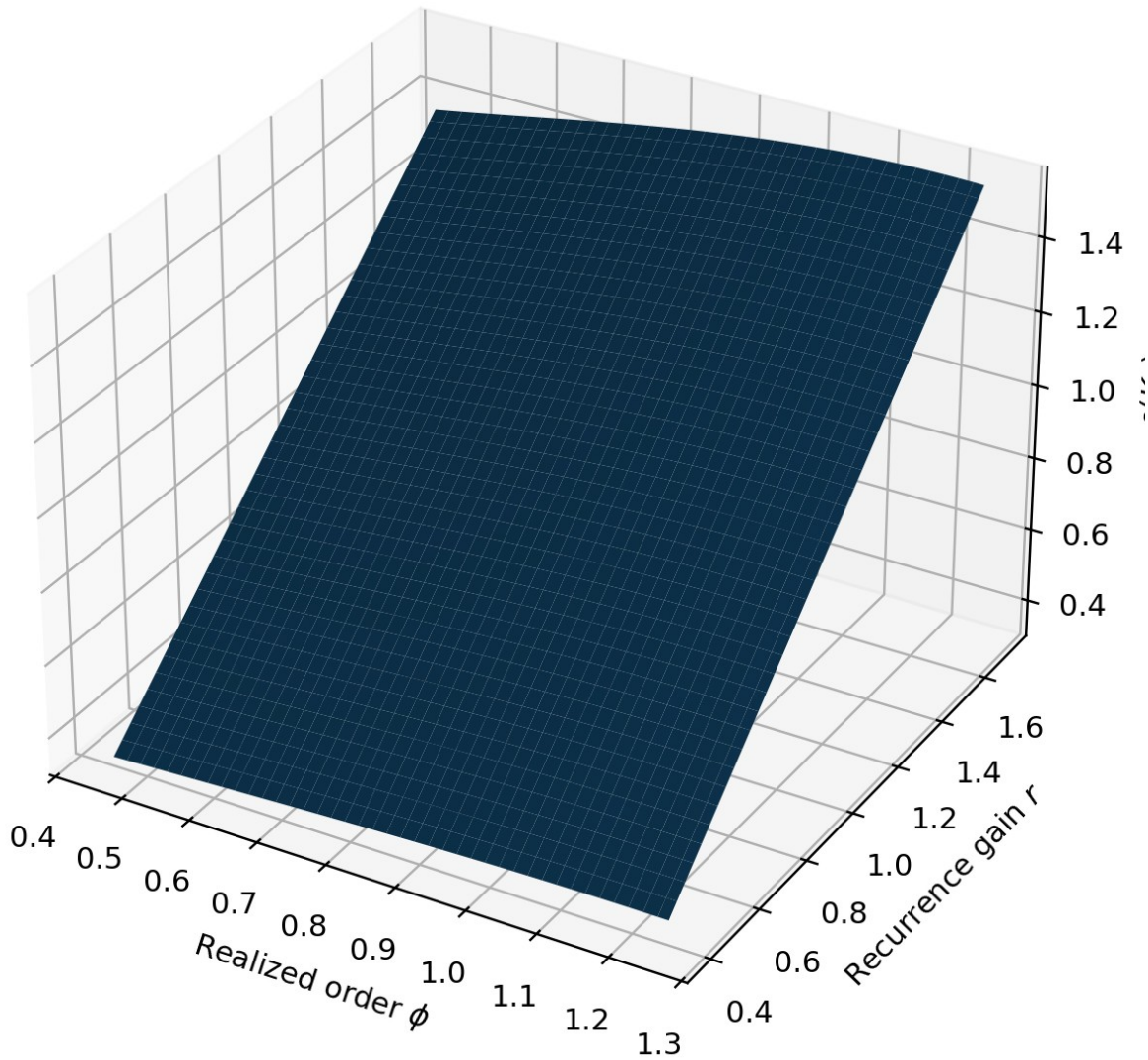
### 16.5 Recursive critical surface and derivative identity

At the benchmark  $\phi^*=0.7481$  and  $g=0.8$ , the previous parameterization [10] uses  $R_0=1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$ . The present execution gives:

$$K_\phi(r,g)=rR_0C_\phi(g) \quad (86)$$

$\rho(R_0C_\phi)=0.8711$ , so  $r_{\text{crit}}=1/\rho=1.1480$ . Choosing  $r=1.1422$  yields  $\rho(K_\phi)=0.995000$ . The analytic resolvent derivative  $\partial_\phi C_\phi = -C_\phi Y'_\phi C_\phi$  was checked against a centered  $h=10^{-6}$  finite difference with relative Frobenius error  $4.297e-10$ .

## Recursive Stability Surface at Fixed Cross-Channel Gain $g=0.8$

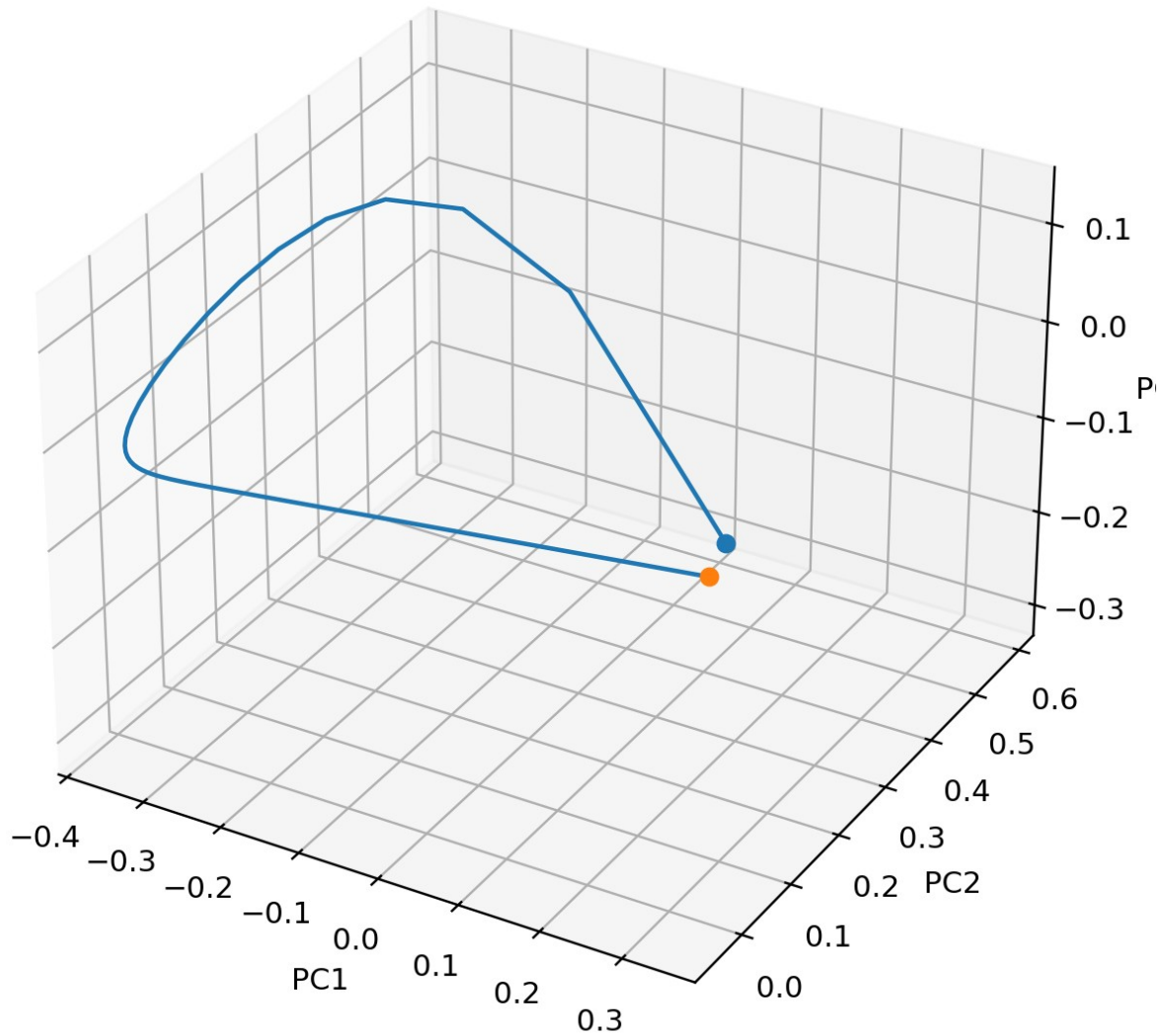


**Figure 7.** Executed 3D recursive stability surface for  $K_\phi = rR_0C_\phi$  at fixed  $g=0.8$ . The critical level  $\rho=1$  separates spectrally subcritical and supercritical regions for this fixed linear model.

### 16.6 Bounded near-critical trajectory

Using the near-critical gain above and the deterministic initial state  $s_0=(0.90,-0.55,0.35,0.25,-0.20)$ , 160 iterations remained bounded. The initial Euclidean norm was 1.1565, the maximum norm over the run was 1.3535, and the final norm was 0.6575. The trajectory is shown in the first three principal coordinates of the 161 simulated states.

## Bounded Near-Critical Recursive Trajectory (PCA Projection)



**Figure 8.** Executed bounded near-critical trajectory projected into three principal coordinates. The run demonstrates power-bounded recurrent behavior for the chosen fixed linear model. It does not demonstrate phenomenal consciousness.

## 17. Interpretation of the simulations

The simulations support five internal statements and only those statements. First, the neutral resolvent gives an exact complementary partition to numerical precision. Second, the canonical spectral gate can be visualized as a clean sign surface without naming any carrier. Third, local order and recursive gain are distinct control coordinates in the declared toy model. Fourth, shared-neutral integration and preserved differentiation compete rather than collapse into a single monotone 'consciousness score'. Fifth, a near-critical recurrent closure can remain bounded under the declared fixed linear dynamics.

None of these results establishes that the URE produces real atoms, protocells, organisms, or qualia. Their function is diagnostic: they show that the unified operator architecture is computationally coherent enough to expose stability corridors, ablation directions, and exact identities that can be preregistered before empirical tests.

## 18. Falsification architecture

A universal theory becomes weaker, not stronger, if every failure can be repaired by redefining a stage. The URE therefore requires explicit failure surfaces. The following failures target the unification itself rather than any one application.

**Table 4.** Primary falsification surfaces of the URE program.

| Failure surface            | Decision condition                                                                                                           |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Formation failure          | $\lambda_0 \geq 0$ or the Level-0 negative mode is not isolated.                                                             |
| Positive-geometry failure  | $\lambda_{\min}(H_C) = 0$ or the master load ceases to be positive on the declared domain.                                   |
| Neutral-resolution failure | Empirical response requires an independent pole or nonlinear law not derivable from the universal $C_N/B_N$ family.          |
| Persistence failure        | The declared witness block is not power bounded or moves only under target-aware spectral selection.                         |
| Generativity failure       | Higher stages can only be obtained by stage-specific $G_n$ maps or outcome-aware thresholds.                                 |
| Carrier-transfer failure   | The frozen role extraction must be redefined after entering a new physical/biological domain.                                |
| Hierarchy-only null        | A capacity-matched generic hierarchy predicts the same held-out transitions without BFG mediation/return structure.          |
| Atomic failure             | The proposed atomic readout adds no predictive or organizing content beyond relabeling established carrier theory.           |
| Abiogenesis failure        | Repeated-cycle closure does not depend on the predicted coupling, export, or witness relations under ablation.               |
| Organism failure           | Projection/readout/mediation variables add no prospective value beyond matched developmental nulls.                          |
| Consciousness failure      | Matched recurrent or connectivity models reproduce held-out perturbational predictions without shared-neutral constraints.   |
| Qualia failure             | No non-circular content-specific closure invariant is found, or local carrier activity fully explains content and ownership. |
| Global no-retuning failure | One domain can be saved only by altering the universal response family or generator after seeing the target data.            |

## 19. Preregistration and no-retuning protocol

Every empirical test of the URE should freeze the following items before target outcomes are inspected: the closure-level partition; the carrier-to-BFG role extraction; the positive load estimator; the neutral response family; the persistent-witness contour or subspace estimator; the stage predicate; the spectral gate threshold or exact sign rule; the return/witness metric; all ablations; and all capacity-matched nulls. The same definitions must be transported to validation data without post-hoc semantic repair.

- Freeze operator family and numerical tolerances before outcome access.
- Separate structural level  $n$  from physical/developmental time  $t$ .
- Report retention, complement, export, spectral margin, and witness continuity jointly.
- Include hierarchy-only, order-independent, return-ablated, shuffled-coupling, and capacity-matched recurrent controls where applicable.
- Preserve negative outcomes and localize the failed bridge instead of rewriting the stage definition.

- For consciousness and qualia, preregister report-independent or report-minimized estimators so that the target labels do not define the claimed invariants.
- For abiogenesis, require repeated-cycle persistence rather than a one-off successful compartment.
- For cross-domain claims, keep the role map fixed and allow carrier units to change only through declared empirical readouts.

## 20. Relationship to established science

The URE is a structural unification proposal, not a replacement for established domain theories. Matrix analysis and perturbation theory constrain the operator statements [7,8]. Fixed-point theory constrains order-indexed self-consistency [9]. Relativistic field theory constrains the physical spin-2 branch [11–15]. Information theory constrains the meaning of information variables [44]. Dissipativity and self-organization theory constrain storage/export language [45–47]. Emergence and hierarchy research provide comparison frameworks for multilevel organization [48,49]. Developmental biology supplies carrier mechanisms for pattern formation [26–30]. Origin-of-life chemistry supplies experimental carrier constraints [17–23]. Consciousness science supplies theory comparisons and perturbational benchmarks [35–41,43].

The scientific burden on BFG is therefore incremental: it must show that the frozen neutral-closure architecture predicts, compresses, or explains held-out structure beyond what these mature frameworks already provide. Similarity of vocabulary is insufficient.

## 21. Limitations and open proof obligations

The new unification construction closes one specific formal gap—the existence of a single level-independent generating map—but several stronger claims remain open.

**Table 5.** Open proof obligations after formal unification.

| Open obligation               | Current status                                                                                                                                                   |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microscopic source uniqueness | The Level-0 operators $\mathcal{D}_0$ , $\mathcal{C}_0$ , and $\mathcal{N}_0$ are not uniquely derived from a deeper empirically established microscopic theory. |
| Cross-carrier representation  | The direct-sum candidate packet is a carrier-neutral construction; a unique physical representation map for every domain is not yet derived.                     |
| Stage existence               | The URE defines how a stage would be generated and classified; it does not yet prove that every named stage necessarily occurs for generic initial data.         |
| Atomic numerics               | Actual atomic spectra and constants are not numerically derived from URE alone; established quantum carrier laws remain necessary empirical constraints.         |
| Abiogenesis history           | The exact historical path on early Earth is not reconstructed.                                                                                                   |
| Morphogenesis                 | Gene regulation, mechanics, morphogens, bioelectricity, and extracellular matrix dynamics are not derived away; they are candidate carrier realizations.         |
| Consciousness identification  | The mathematical shared-neutral closure theorem does not by itself prove that the regime is phenomenal consciousness.                                            |
| Qualia identification         | The URE equivalence class removes a free formal $\text{Inv}_G$ , but empirical mapping of classes to reported qualitative content remains open.                  |
| Thermodynamic units           | Energy and entropy readouts require domain-specific constitutive measurement maps.                                                                               |
| Higher collective closure     | Narrative, intersubjective, scientific, and civilizational stages are substantially more speculative than the physical and biological core.                      |

## 22. Research program

The immediate research program should proceed in the order imposed by the URE rather than by application novelty. First, derive or constrain the next-level formation operator  $K_{\{n+1\}}$  directly from the retained/persistent-complement packet in explicit finite-dimensional models and test robustness under perturbation. Second, construct at least one carrier-realized atomic-to-molecular case in which the BFG load and witness projector are estimated from established quantum data without changing the universal response family. Third, perform a preregistered prebiotic systems-chemistry test with repeated cycles and hierarchy-only nulls. Fourth, transport the same estimator family into an organoid or developmental dataset with frozen role definitions. Fifth, test the shared-neutral cognitive branch against capacity-matched recurrent models in perturbational data. Sixth, test content-specific closure equivalence on controlled perceptual data before making any stronger qualia claim.

The key success criterion is not the number of domains in which BFG language can be applied. It is the number of domains that survive the same operator family, the same gate logic, and the same failure accounting without hidden retuning.

## 23. Universal Reclosure Theorem Candidate

| Theorem                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | candidate | — | Universal | Reclosure |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|---|-----------|-----------|
| Assume: (i) a simple negative Level-0 formation mode; (ii) positive master closure geometry at every admitted iterate; (iii) exact neutral elimination with $C_N+B_N=I$ ; (iv) a spectrally defined power-bounded witness subspace; (v) reapplication of the same $H_C$ , $L_C$ , neutral-response family, and formation operator construction to the retained/persistent-complement packet; and (vi) the common spectral gate $g_U=1$ iff $\lambda_{\min}(K_{\text{next}})<0$ . Then every admitted higher closure is generated by the single recursion $X_{\{n+1\}}=\mathcal{U}[X_n]$ . If stage labels are fixed invariant predicates on the resulting iterates, no stage requires an independent fundamental BFG response law. This conclusion is formal and does not imply empirical realization of the assumptions. |           |   |           |           |

Proof sketch. Exact neutral elimination supplies the complementary pair and the retained/complement budget. The witness projector selects the power-bounded recursive block. The candidate next-level packet is formed only from these upstream outputs. Positive master geometry is reevaluated on that packet using the same operator family. The same spectral formation criterion either admits or terminates the next closure. Induction therefore shows that every admitted iterate is a composition of one fixed map  $\mathcal{U}$ . Because stage predicates act only after an iterate is generated, they cannot alter the dynamics. Any empirical application that requires modifying  $\mathcal{U}$  therefore lies outside the theorem candidate and counts as no-retuning failure.

## 24. Conclusion

This paper turns the BFG cross-domain program into a single explicit no-retuning recursion. Level-0 formation provides the initial polarity branch. Exact neutral elimination provides the universal retained/complementary pair. Positive master geometry supplies the common load. Recursive conjugate closure supplies persistence. Local fixed-point order and witness continuity supply the recursive order constraints. The canonical spectral gate and LPCP supply the generativity criterion and the local physical bridge. The new Universal Reclosure Equation reuses those same ingredients at every admitted level.

Within that formal architecture, atom, molecule, protocell, cell, organism, cognition, conscious subjectivity, qualitative content, and self-owned experience are no longer introduced by separate BFG fundamental laws. They are frozen invariant classes of the same URE orbit. The executed simulations show that the exact neutral partition, spectral gating, order–feedback stability, shared-neutral integration/differentiation trade-off, and bounded near-critical recurrence are computationally coherent in the declared models.

The resulting claim is deliberately narrow enough to be falsifiable: one generating architecture, one neutral response family, one persistence logic, one spectral gate, and no hidden downstream retuning. The next step is not to add more sectors. It is to calculate and test the same frozen structure on increasingly demanding carrier data until the unification either survives or localizes its first genuine failure.

## Appendix A. Core formula register

**Table 6.** Compact formula register.

| Role                   | Formula / statement                                                                            |
|------------------------|------------------------------------------------------------------------------------------------|
| Level-0 source         | $\mathcal{T}_0 = (\mathcal{D}_0, c_0, \mathbb{N}_0); K_0 = c_0 + \mathbb{N}_0 - \mathcal{D}_0$ |
| Formation              | $\delta^*_{\pm} = \pm \sqrt{(-\lambda_0) c_0}$ for simple $\lambda_0 < 0$                      |
| Positive load          | $Y = A \dagger A \geq 0; Y\_C = L\_C \dagger H\_CL\_C$                                         |
| Neutral retention      | $C\_N(Y) = (I + Y)^{-1}$                                                                       |
| Complement             | $B\_N(Y) = Y(I + Y)^{-1} = I - C\_N(Y)$                                                        |
| Exact partition        | $C\_N + B\_N = I$ ; quadratic retained+complement budget exact                                 |
| Persistence            | $R\_Ce\_T = e\_T$ ; stable complement strictly contractive in a Lyapunov norm                  |
| Order fixed point      | $\varphi\_nv\_n(\varphi\_n) = \Phi\_src, n$                                                    |
| Witness subspace       | $\mathcal{W}\_n = \{z: \sup_k  R\_C, n^k z  < \infty\}$                                        |
| Keep / upward / export | $P\_WC\_ND; P\_WB\_ND; (I - P\_W)B\_ND$                                                        |
| Candidate packet       | $\tilde{V}D\_{\{n+1\}} = D\_keep \oplus D\_up$                                                 |
| Spectral gate          | $g\_U = 1$ iff $\lambda\_min(K\_{\{n+1\}}) < 0$                                                |
| URE                    | $X\_{\{n+1\}} = g\_U C\_N(Y\_{\{n+1\}})[D\_keep \oplus D\_up]$                                 |
| Iteration              | $X\_n = \mathcal{U}^n[X_0]$ before terminal level $N\_*$                                       |
| Cognitive load         | $Y\_ \varphi = M\_ \varphi \dagger M\_ \varphi$                                                |
| Cognitive integration  | $\Xi\_ \varphi = \sum_{i \neq j}  P\_i C\_ \varphi P\_j ^2_{HS} /  C\_ \varphi ^2_{HS}$        |
| Differentiation        | $\Delta\_ \varphi = \sigma\_min(C\_ \varphi   H\_act)$                                         |
| Qualia equivalence     | $Q[X] = [X]\_{\sim BFG}$                                                                       |

## Appendix B. Simulation reproducibility register

**Table 7.** Exact numerical specification for executed figures.

| Item                      | Specification                                                                                                                           |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Software                  | Python 3.13.5; NumPy 2.3.5; Matplotlib 3.10.8.                                                                                          |
| Neutral surface           | $y_1, y_2$ logarithmically sampled from $10^{-2}$ to $10^2$ ; $160 \times 160$ grid; $\Gamma\_rc = \frac{1}{2}(4c_1b_1 + 4c_2b_2)$ .    |
| Order-feedback surface    | $\varphi \in [0.45, 1.55]$ , $B \in [0.55, 0.98]$ ; $\alpha = 0.35$ , $\beta = 0.25$ , $d = 0.45$ , $c = 0.08$ ; $180 \times 180$ grid. |
| Spectral gate normal form | $\delta \in [-1.6, 1.6]$ , $g \in [0, 1.6]$ ; $\lambda\_min = 1 - \sqrt{(\delta^2 + g^2)}$ ; $180 \times 180$ grid.                     |
| Cognitive W               | Unit-weight ring edges (1,2),(2,3),(3,4),(4,5),(5,1); chords $W13 = 0.35$ , $W24 = 0.25$ , $W35 = 0.30$ ; normalized to $ W _2 = 1$ .   |
| Cognitive D               | diag(0.55, 0.65, 0.75, 0.85, 0.95).                                                                                                     |
| Order fixed point         | $\Phi\_src = 1$ ; $a = 0.45$ ; $\varphi^* = 0.7481333923$ .                                                                             |
| Integration surface       | $\varphi \in [0.45, 1.25]$ , $g \in [0, 1.8]$ ; $140 \times 140$ grid.                                                                  |

|                     |                                                                                                                      |
|---------------------|----------------------------------------------------------------------------------------------------------------------|
| Recursive primitive | $R_0 = 1 \oplus 0.96R(0.23) \oplus 0.86R(-0.37)$ .                                                                   |
| Near-critical gain  | $r = 0.995/\rho(R_0 C_\varphi) = 1.1422297734$ at $\varphi^* = 0.7481333923$ , $g = 0.8$ .                           |
| Derivative check    | Centered finite difference $h = 10^{-6}$ ; relative Frobenius error $4.2967 \times 10^{-10}$ .                       |
| Trajectory          | Initial $s_0 = (0.90, -0.55, 0.35, 0.25, -0.20)$ ; 160 iterations; PCA obtained by SVD of centered 161-state matrix. |

## Appendix C. Source-credit register

**Table 8.** Source-line provenance used by the Unification Paper.

| Source line                                             | Use in present paper                                                                                                                        |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Marcel Theodor Wendé & Thomas W. Loker [3]              | Co-authored Level-0 admissibility and pre-polar formation source line.                                                                      |
| Marcel Theodor Wendé & Luke Kenneth Casson Leighton [4] | Recursive conjugate-closure source line $R_C = MRM^\dagger$ , $M = \sqrt{N}$ and related stability geometry.                                |
| Canonical BFG source [1]                                | Universal positive closure architecture, exact neutral elimination, spectral gate, and LPCP.                                                |
| Prior BFG sources [5,6]                                 | Local order, fixed-point mediation, generativity/return architecture, and cross-scale compatibility.                                        |
| Prior carrier-specific BFG sources [10,16,24,25,33,42]  | Carrier-level structural constraints for atomic, biological, organismic, cognitive, phenomenal, and thermodynamic readouts.                 |
| This paper                                              | Universal Reclosure Equation, level-independent generativity construction, stage-predicate formulation, and executed synthesis simulations. |

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## Declarations

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Author contribution. Conceptualization, BFG unification architecture, theoretical formulation, interpretation, and manuscript responsibility: Marcel Theodor Wende. The Level-0 and recursive conjugate-closure source lines used in the manuscript are credited through the numbered bibliography.

Data availability. No external empirical dataset was analyzed. Numerical results are synthetic deterministic calculations from equations and parameters fully stated in the manuscript.

Code availability. The simulation equations, parameter ranges, software versions, and deterministic initial conditions required to reproduce the displayed figures are provided in Appendix B. A public repository release should include the exact script and environment lockfile before journal submission.

Ethics. No human participants, animal experiments, or identifiable personal data were used.

Competing interests. No competing-interest statement is asserted beyond the author's responsibility for the BFG research program; this item should be updated to the target journal's required form before submission.

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